CHE 451 Renewable Energy Technologies

2. **<u>Credits, contact hours and categorization</u>**
   Credit hours: 3; Contact hours: One, 165 min/week lecture; Engineering

3. **<u>Instructor's or course coordinator's name</u>**
   Said Al-Hallaj

4. **<u>Text book, title, author, publisher, year</u>**
   "Hybrid Hydrogen Systems for Stationary and Transportation Applications", Said Al-Hallaj and Kristofer Kiszynski, 1st Edition, 2011, Springer Publishing, ISBN 978-1-84628-466-3.

5. **<u>Specific course information</u>**
   a. <u>Brief description of the content of the course (catalog description)</u>
      The course will cover fundamentals of renewable energy technologies such as solar, wind and biomass. In addition, course will provide an introduction to various energy storage technologies such as batteries and fuel cells. Finally, the hydrogen economy promises and challenges such as hydrogen storage and transportation will be discussed and analyzed.
   b. <u>Prerequisites or co-requisites</u>
      Consent of the instructor.
   c. <u>Indicate whether a required, elective, or selected elective.</u>
      Elective.

6. **Specific goals for the course**
   a. <u>Specific outcomes of instruction</u>. Students will be able to:
      - Understand the fundamentals and essential concepts used in renewable energy technologies such as solar, wind, biomass.
      - Understand and analyze energy storage technologies such as batteries, fuel cells and supercapacitors.
      - Address key ideas, as well as technical and economic issues related to renewable energy in a final project term paper.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcomes	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.			X	
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.				X
(3) an ability to communicate effectively with a range of audiences.				X
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.				X
(5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.				X
(6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	X			
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.				X

7. Brief list of topics to be covered

- Introduction (Renewable Energy and Sustainability)
- Solar Energy Calculations and Solar Photovoltaics (PV)
- Building Integrated Photovoltaics
- Wind Energy Basics and Applications
- Bioenergy
- Hydrogen Economy and Hydrogen Storage
- Fuel Cells
- Energy Storage for Smart Grid Applications
- Geothermal Energy
- Hydrogen Economy: Production and Infrastructure
- Optimization of Renewable Hydrogen Energy Systems
- Control of Hybrid Fuel Cell/Battery Systems
- Solar Desalination and Hybrid Fuel Cell/Desalination